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Functional equations

Level 4 Olympiad · Algebra

How to find every function satisfying an equation: smart substitutions, injectivity and surjectivity, Cauchy's equation over the rationals, and complete write-ups that check the answer.

By the end you can: You will be able to choose substitutions that pin down special values and make awkward terms vanish; You will be able to prove and use injectivity and surjectivity, and solve Cauchy's equation over the rationals; You will be able to write a complete 'find all functions' solution, avoiding the pointwise trap and checking every answer.

Lesson

Let us start with a problem. Find every function $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x) + 2f(1 - x) = x^2 \quad \text{for all real } x.$$

The equation ties the value at x to the value at $1 - x$, so on its own it does not tell us $f(x)$. But it holds for *every* real number, so it also holds with x replaced by $1 - x$:

$$f(1 - x) + 2f(x) = (1 - x)^2.$$

Now we have two linear equations in the two unknowns $f(x)$ and $f(1 - x)$. Doubling the second and subtracting the first gives $3f(x) = 2(1 - x)^2 - x^2$, so

$$f(x) = \frac{x^2 - 4x + 2}{3}.$$

We are not finished. So far we have shown only that *if* a solution exists, it must be this function. Substituting back, $f(1 - x) = \frac{1}{3}(x^2 + 2x - 1)$ and $f(x) + 2f(1 - x) = \frac{1}{3}(3x^2) = x^2$, so it really is a solution, and it is the only one. That two-part shape, "any solution must be this" followed by "and this works", is the skeleton of every functional equation you will ever write up.

What "find all functions" asks for

A complete answer has two halves. **Necessity:** assuming f satisfies the equation, deduce that f is one of an explicit list. **Sufficiency:** check that every function on the list satisfies the equation for *all* allowed values of the variables. At BMO a missing check costs marks, and occasionally a candidate you derived fails the check, which is exactly why it is required.

The domain is part of the problem. The same formula can have different answers on $\mathbb{R}$, on $\mathbb{Q}$ and on the positive integers, and a substitution like $x = 0$ is illegal if 0 is not in the domain. Read the first line of the problem twice.

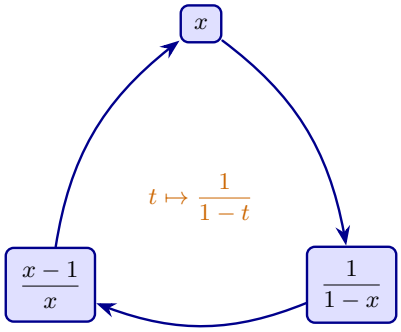
The first substitutions

An equation that holds for all x and y is a machine that produces one fact for every choice of inputs. Your job is to feed it inputs that produce useful facts. The standard first moves are these.

- Set variables to 0 (or to 1 in a multiplicative equation) to find $f(0)$ or $f(1)$.
- Try $y = x$ and $y = -x$; if the equation is not symmetric, swap x and y and compare.

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- Choose one variable in terms of the other so that an awkward argument becomes a value you know, or so that two arguments become equal.
- Replace x by an expression whose repeated application cycles back to x, such as $1-x$ (a cycle of length two, as in the opening problem) or $\frac{1}{1-x}$ (a cycle of length three).



With a cycle of length three you substitute twice and get three equations in three unknowns, which you then solve exactly as above.

Example 1. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x-y) = f(x) + f(y) - 2xy$ for all real x, y.

Solution.

The term $-2xy$ looks like part of $(x-y)^2$, so a fair guess is $f(x) = x^2$. Now look for a substitution that turns the awkward term $f(x-y)$ into something known. Putting $y = x$ makes the argument 0, so first we need $f(0)$.

With $x = y = 0$ the equation reads $f(0) = 2f(0)$, so $f(0) = 0$. With $y = x$ it reads $f(0) = 2f(x) - 2x^2$, so $f(x) = x^2$ for every real x. That is necessity. For sufficiency, $(x-y)^2 = x^2 + y^2 - 2xy$ for all x, y, so $f(x) = x^2$ is the unique solution.

Notice how little work this took once we asked the right question: which substitution makes an argument constant?

Injective, surjective, and making terms vanish

A function is **injective** if $f(a) = f(b)$ forces $a = b$, and **surjective** if every value in the codomain is taken. Both are proved from the equation, never assumed. The most common source is a relation like $f(f(x)) = x + c$: if $f(a) = f(b)$ then applying f gives $a + c = b + c$, so f is injective, and since $x + c$ takes every real value, f is surjective.

Injectivity lets you strip an f off both sides: if you can arrange $f(A) = f(B)$, then $A = B$. Surjectivity lets you replace "for all y, a statement about $f(y)$" by "for all real t, the same statement about t".

Example 2. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f\big(xf(y) + y\big) = xy + f(y)$ for all real x, y.

Solution.

Guessing $f(x) = ax + b$ and comparing coefficients gives $a^2 = 1$ and $b = 0$, so we expect $f(x) = x$ and $f(x) = -x$. Now we must prove nothing else works.

Put $y = 0$: $f\big(xf(0)\big) = f(0)$ for all x. If $f(0) = c \neq 0$, then xc runs through every real number, so f is constant, equal to c; but then the equation says $c = xy + c$ for all x, y, which is false. Hence $f(0) = 0$.

A natural next idea is to make the left-hand argument vanish: if $f(y) \neq 0$, take $x = -y/f(y)$ to get $0 = -y^2/f(y) + f(y)$, so $f(y)^2 = y^2$. It is tempting to stop here and announce $f(y) = \pm y$.

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That would be wrong: we only know that *for each* y the value is y or $-y$, and the sign could change from one y to another. We need a better idea.
First, f has no zeros except 0: if $f(b)=0$, then putting $y=b$ gives $f(b)=xb+f(b)$ for all x, so $b=0$. In particular $k=f(1)\neq 0$. Now fix $y=1$:

$$f(kx+1)=x+k \quad \text{for all real } x.$$

Since $k\neq 0$, as x runs over $\mathbb{R}$ so does $t=kx+1$, and $x=(t-1)/k$. Hence $f(t)=\frac{t-1}{k}+k$ for every real t: the function is linear. Now $f(0)=0$ gives $k^2=1$, so $f(t)=t$ or $f(t)=-t$.
Both work: for $f(t)=t$ each side is $xy+y$, and for $f(t)=-t$ the left side is $f(-xy+y)=xy-y$, which equals the right side $xy-y$. The moral: fixing one variable can make the equation describe f along a whole line, and if that line covers the domain, you have f.

Cauchy's equation over the rationals

Theorem (Cauchy over $\mathbb{Q}$). If $f:\mathbb{Q}\to\mathbb{R}$ satisfies $f(x+y)=f(x)+f(y)$ for all rationals x,y, then $f(x)=cx$ for all rational x, where $c=f(1)$.
Proof. Putting $x=y=0$ gives $f(0)=0$, and $y=-x$ gives $f(-x)=-f(x)$. Since $f\big((n+1)x\big)=f(nx)+f(x)$, induction gives $f(nx)=nf(x)$ for every positive integer n. So $f(m)=mc$ for integers m, and for $n>0$ we have $nf\big(\frac{m}{n}\big)=f(m)=mc$, so $f\big(\frac{m}{n}\big)=c\cdot\frac{m}{n}$.
On $\mathbb{R}$. Additivity alone is not enough: there exist wild additive functions on $\mathbb{R}$ (built using the axiom of choice) that are not of the form cx. You need one extra fact. A convenient one: if f is additive on $\mathbb{R}$ and $f(x)\geq 0$ whenever $x\geq 0$, then $f(x)=cx$. Indeed, for $a<b$ we get $f(b)=f(a)+f(b-a)\geq f(a)$, so f is increasing; for real x and rationals $p<x<q$ we get $cp\leq f(x)\leq cq$, and letting $p,q\to x$ gives $f(x)=cx$. Monotonicity, continuity, or boundedness on an interval would also do.

Example 3. Find all functions $f:\mathbb{Q}\to\mathbb{Q}$ such that $f\big(x+f(y)\big)=f(x)+4y$ for all rationals x,y.
Solution.
Trying $f(x)=kx$ gives $k^2=4$, so we expect $f(x)=2x$ and $f(x)=-2x$. There is no obvious substitution that computes f directly, so we first collect structure.
Put $x=0$: $f\big(f(y)\big)=c+4y$, where $c=f(0)$. The right-hand side takes each rational value exactly once as y varies, so f is injective and surjective. Put $y=0$: $f(x+c)=f(x)$, and injectivity gives $x+c=x$, so $c=0$ and $f\big(f(y)\big)=4y$.
Now we aim for additivity. The original equation contains $f(y)$ inside; replace y by $f(y)$ to get $f\big(x+f(f(y))\big)=f(x)+4f(y)$, that is,

$$f(x+4y)=f(x)+4f(y).$$

Putting $x=0$ gives $f(4y)=4f(y)$, so $f(x+4y)=f(x)+f(4y)$. Since $4y$ is an arbitrary rational, f is additive, and Cauchy over $\mathbb{Q}$ gives $f(x)=kx$ with $k=f(1)$ rational. Then $4y=f(f(y))=k^2y$, so $k=\pm 2$.
Check: if $f(x)=2x$, the left side is $2x+4y$; if $f(x)=-2x$, the left side is $f(x-2y)=-2x+4y$. Both equal the right side. If the 4 were replaced by 2, the same argument would force $k^2=2$, which has no rational solution, so there would be no solutions at all on $\mathbb{Q}$.

Common traps.

- **The pointwise trap.** Showing that for each x, $f(x)=x$ or $f(x)=-x$, does not show

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that f is one of those two functions. You must rule out mixtures.

- **No final check,** or a check with a few special values only. The check must be for all values.
- **Dividing by something that might be zero,** such as $f(y)$ or x. Deal with the zero case separately, as in Example 2.
- **Cauchy on $\mathbb{R}$ without justification.** Additivity on $\mathbb{R}$ gives cx only with an extra condition, which you must prove.
- **Assuming the form.** "Let $f(x) = ax + b$ and compare coefficients" finds candidates; it proves nothing about other functions.
- **Ignoring the domain,** for example substituting 0 when f is defined only on positive integers.

How to spot this in a contest

At BMO Round 1 a functional equation usually appears as a "find all functions" problem near the end of the paper, typically with one or two solutions, often linear. A good routine: guess the answers first (constants and $ax + b$), since knowing the target tells you which facts to aim for; find $f(0)$, $f(1)$ and other special values; look for an $f(f(x))$ relation, which signals injectivity or surjectivity; try to make arguments equal or make them vanish; if additivity appears, check the domain before quoting Cauchy. On integer domains, think induction, parity and size.

In the AIME the function is usually given by a few rules and you must compute one value. The same substitutions apply, together with spotting a period or a familiar function hiding behind the rules, but remember that a guessed function is only safe if the rules force it.

Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆ Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) + x\,f(1 - x) = 2x - x^2$ for all real x.

2. ⋆⋆ A function f is defined for all real x other than 0 and 1, and satisfies $f(x) + f\left(\dfrac{1}{1-x}\right) = x$ for all such x. Find $f(3)$.

3. ⋆⋆ Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that $f(x + y) = f(x) + f(y) + xy(x + y)$ for all rationals x, y.

4. ⋆⋆⋆ Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\big(xf(y) + f(x)\big) = f(x)f(y) + x$$

for all real x, y.

5. ⋆⋆⋆ Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that $f\big(f(n)\big) = n + 2$ for every integer n.

6. ⋆⋆⋆⋆ Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 + y) = f(x)^2 + f(y)$$

for all real x, y.

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Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. AIME 1984 Problem 7 ⋆⋆
2. AIME 1984 Problem 12 ⋆⋆
3. AIME 1988 Problem 8 ⋆⋆
4. AIME 1997 Problem 12 ⋆⋆⋆
5. BMO Round 1 2001/02 Problem 5 ⋆⋆⋆
6. BMO Round 1 2009/10 Problem 5 ⋆⋆⋆
7. BMO Round 1 2007/08 Problem 6 ⋆⋆⋆
8. BMO Round 1 2019/20 Problem 6 ⋆⋆⋆⋆
9. BMO Round 1 2014/15 Problem 6 ⋆⋆⋆⋆
10. USAMO 2002 Problem 4 ⋆⋆⋆⋆

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Hints

1. Replace x by $1-x$ and solve the two linear equations. Before you divide, check that the coefficient can never be zero.
2. Apply the substitution $x \mapsto \frac{1}{1-x}$ to 3, then to the result, then again. Where do you end up?
3. Find one solution of the form kx^3, subtract it, and see what equation the difference satisfies.
4. Show that $f(0) = 0$; then $y = 0$ tells you something strong about $f(f(x))$. After that, fix $x = 1$.
5. Work out $f\big(f(f(n))\big)$ in two different ways. Then the values $f(0)$ and $f(1)$ determine everything, and parity matters.
6. Find $f(0)$, show that f is odd and then additive. To finish, ask what sign f takes on non-negative numbers.

Solutions

Problem 1

Answer: $f(x) = x$

Replacing x by $1-x$ gives $f(1-x)+(1-x)f(x) = 2(1-x)-(1-x)^2 = 1-x^2$, so $f(1-x) = 1-x^2-(1-x)f(x)$. Substituting this into the original equation gives $f(x)+x(1-x^2)-x(1-x)f(x) = 2x-x^2$, that is,

$$(x^2 - x + 1)\, f(x) = x^3 - x^2 + x = x(x^2 - x + 1).$$

Since $x^2 - x + 1 = \left(x - \frac{1}{2}\right)^2 + \frac{3}{4} > 0$ for every real x, we may divide to get $f(x) = x$. Conversely $f(x) = x$ gives $x + x(1 - x) = 2x - x^2$, so it is a solution, and it is the only one.

Problem 2

Answer: $\frac{25}{12}$

The map $t \mapsto \frac{1}{1-t}$ sends $3 \mapsto -\frac{1}{2} \mapsto \frac{2}{3} \mapsto 3$, and none of these numbers is 0 or 1. Writing $A = f(3)$, $B = f\left(-\frac{1}{2}\right)$ and $C = f\left(\frac{2}{3}\right)$, the equation at $x = 3$, $x = -\frac{1}{2}$ and $x = \frac{2}{3}$ gives

$$A + B = 3, \qquad B + C = -\tfrac{1}{2}, \qquad C + A = \tfrac{2}{3}.$$

Adding the first and third and subtracting the second gives $2A = 3 + \frac{1}{2} + \frac{2}{3} = \frac{25}{6}$, so $f(3) = \frac{25}{12}$.

Such a function does exist, so the question makes sense: with $u = \frac{1}{1-x}$ and $w = \frac{x-1}{x}$, the function $f(x) = \frac{1}{2}(x - u + w)$ satisfies $f(x) + f(u) = \frac{1}{2}(x - u + w) + \frac{1}{2}(u - w + x) = x$, because the map sends u to w and w to x.

Problem 3

Answer: $f(x) = \frac{1}{3}x^3 + cx$ for a rational constant c

Since $(x+y)^3 = x^3 + y^3 + 3xy(x+y)$, the function $\frac{1}{3}x^3$ satisfies the equation. Let $g(x) = f(x) - \frac{1}{3}x^3$, which also maps $\mathbb{Q}$ to $\mathbb{Q}$. Subtracting the identity for $\frac{1}{3}x^3$ from the given equation shows that $g(x + y) = g(x) + g(y)$ for all rationals x, y. By Cauchy's theorem over $\mathbb{Q}$, $g(x) = cx$ where $c = g(1)$ is rational. Hence $f(x) = \frac{1}{3}x^3 + cx$.

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Conversely, for any rational c this f maps rationals to rationals, and $f(x+y)-f(x)-f(y)=\frac{1}{3}\left[(x+y)^3-x^3-y^3\right]=xy(x+y)$, so every such function is a solution.

Problem 4

The solutions are $f(x)=x$ and $f(x)=-x$. Write $P(x,y)$ for the given equation.

$P(0,y)$ gives $f\big(f(0)\big)=f(0)f(y)$ for all y. If $f(0)\neq 0$, then $f(y)=f(f(0))/f(0)$ for all y, so f is a constant k; but then $P(x,y)$ reads $k=k^2+x$ for all real x, which is impossible. Therefore $f(0)=0$.

$P(x,0)$ now gives $f\big(f(x)\big)=x$ for all real x. In particular every real number t is a value of f, since $t=f\big(f(t)\big)$.

Let $a=f(1)$. Then $P(1,y)$ gives $f\big(f(y)+a\big)=af(y)+1$ for all y. Given any real t, choose $y=f(t)$, so that $f(y)=t$; this yields $f(t+a)=at+1$ for every real t. Writing $s=t+a$, we get $f(s)=as-a^2+1$ for all real s. Since $f(0)=0$, we need $a^2=1$, so either $a=1$ and $f(s)=s$, or $a=-1$ and $f(s)=-s$.

Finally we check both. If $f(x)=x$, both sides equal $xy+x$. If $f(x)=-x$, the left side is $f(-xy-x)=xy+x$ and the right side is $(-x)(-y)+x=xy+x$. So both functions are solutions, and there are no others.

Problem 5

The solutions are the functions $f(n)=n+a$ for even n and $f(n)=n+2-a$ for odd n, one for each odd integer a.

Applying f to both sides of $f(f(n))=n+2$ gives $f\big(f(f(n))\big)=f(n+2)$. On the other hand, the equation applied at the integer $f(n)$ gives $f\big(f(f(n))\big)=f(n)+2$. Hence $f(n+2)=f(n)+2$ for all n, and by induction in both directions $f(n+2k)=f(n)+2k$ for all integers n,k.

Let $a=f(0)$ and $b=f(1)$. Then $f(n)=n+a$ for even n and $f(n)=n-1+b$ for odd n. If a were even, we would have $f(a)=2a$, but $f(a)=f(f(0))=2$, giving $a=1$, which is not even. So a is odd, and then $f(a)=a-1+b$; since $f(a)=f(f(0))=2$, we get $b=3-a$. Therefore $f(n)=n+a$ for even n and $f(n)=n+2-a$ for odd n, with a odd.

Conversely, let a be any odd integer and define f this way. If n is even, $f(n)=n+a$ is odd, so $f(f(n))=(n+a)+2-a=n+2$. If n is odd, $f(n)=n+2-a$ is even, so $f(f(n))=(n+2-a)+a=n+2$. So every such function works. For example, $a=1$ gives $f(n)=n+1$, while $a=3$ gives $f(n)=n+3$ for even n and $f(n)=n-1$ for odd n.

Problem 6

The solutions are $f(x)=0$ and $f(x)=x$.

Putting $x=0$ gives $f(y)=f(0)^2+f(y)$, so $f(0)=0$. Putting $y=0$ gives $f(x^2)=f(x)^2$ for all x; in particular $f(t)\geq 0$ for all $t\geq 0$. Putting $y=-x^2$ gives $0=f(x)^2+f(-x^2)$, so $f(-x^2)=-f(x^2)$; that is, $f(-t)=-f(t)$ for every $t\geq 0$, and hence f is odd.

Next we show that f is additive. For $t\geq 0$ and any y, putting $x=\sqrt{t}$ gives $f(t+y)=f(\sqrt{t})^2+f(y)=f(t)+f(y)$. For $t<0$, let $s=-t>0$; applying what we have just proved to s and $t+y$ gives $f(y)=f\big((t+y)+s\big)=f(t+y)+f(s)$, so $f(t+y)=f(y)-f(s)=f(y)+f(t)$ by oddness. So $f(t+y)=f(t)+f(y)$ for all real t,y.

An additive function that is non-negative on $[0,\infty)$ is of the form $f(x)=cx$: for $a<b$ we have $f(b)=f(a)+f(b-a)\geq f(a)$, so f is increasing; additivity gives $f(q)=cq$ for rational q, where $c=f(1)$, and for real x and rationals $p<x<q$ we get $cp\leq f(x)\leq cq$, so letting $p,q\to x$ gives $f(x)=cx$.

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Finally $f(x^2) = f(x)^2$ becomes $cx^2 = c^2x^2$, so $c = 0$ or $c = 1$. Both work: $f(x) = 0$ gives $0 = 0$, and $f(x) = x$ gives $x^2 + y = x^2 + y$.

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